

University of Computer Studies

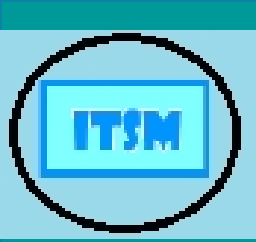


Chapter (5)

Accounting in ERP System

ERP Group

Department of Information Technology
Supporting and Maintenance

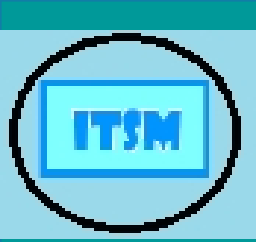


Objectives



After completing this chapter, you will be able to:

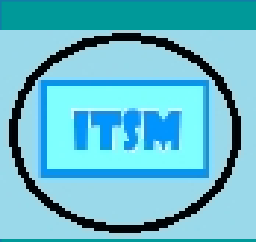
- Describe the differences between financial and managerial accounting
- Identify and describe problems associated with accounting and financial reporting in unintegrated information systems
- Describe how ERP systems can help solve accounting and financial reporting problems in an unintegrated system



Introduction



- In this chapter, you will learn about the activities in the accounting functional area
- Accounting is tightly integrated with all other functional areas
- Accounting activities are necessary for decision making



Accounting Activities

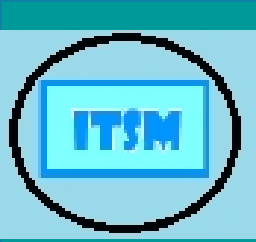


Areas of accounting:

- ❖ Financial accounting
- ❖ Managerial accounting

Financial accounting

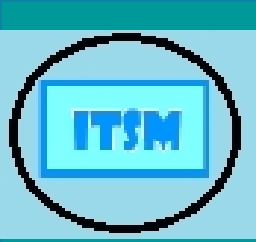
- ❖ Documenting all transactions of a company that have an impact on the financial state of the firm
- ❖ Using documented transactions to create reports for external parties and agencies
- ❖ Reports, or financial statements, must follow prescribed rules and guidelines of various agencies



Accounting Activities (cont'd.)



- Common financial statements: balance sheets and income statements
- Balance sheet
 - Statement that shows account balances such as:
 - ❖ Cash held
 - ❖ Amounts owed to company by customers
 - ❖ Cost of raw materials and finished-goods inventory
 - ❖ Value of fixed assets such as buildings
 - ❖ Amounts owed to vendors, banks, and other creditors
 - ❖ Amounts that the investors have invested in company

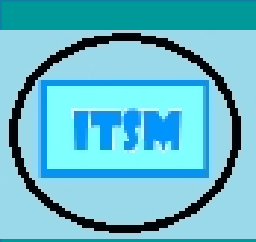


Accounting Activities (cont'd.)



- **Income statement(or) Profit and Loss Statement**

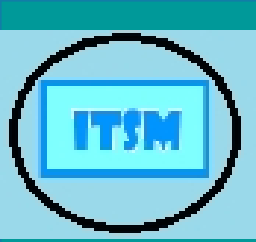
- ❖ Profit and loss (P&L) statement
- ❖ Shows company's revenue, expenses and profit or loss for a period of time (typically a quarter or year)
- ❖ Integrated information system simplifies the process of closing the books and preparing financial statements



Accounting Activities (cont'd.)



- **Quarterly financial statement**
 - ❖ Close books
 - ❖ Closing entries to nominal accounts
 - ❖ Nominal accounts – zero balance to start next cycle
 - ❖ Ensure accounts accurate and up-to-date
 - ❖ “Adjusting” entries
- **Integrated information system advantage**
 - ❖ Simplifies process of closing books and preparing financial statements

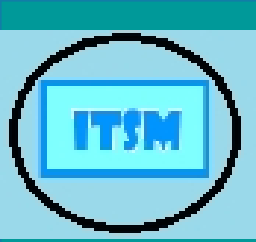


Sales Quotations and Orders



- **Managerial accounting**

- Determine costs and profitability of company's activities
- Provide managers with detailed information
 - ❖ Making information decisions
 - ❖ Creating budgets
 - ❖ Determining the probability
- Information that managers use to control day-to-day activities, develop long-term plans



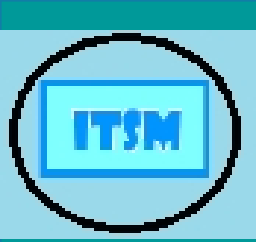
Using ERP for Accounting Information



- Problems associated with unintegrated systems
 - ❖ Data sharing usually did not occur in real time
 - Accounting's data were often out of date
 - ❖ Accounting personnel had to do significant research
- ERP system, with its centralized database, avoids these problems general ledger is record that the company's accounts are kept in.



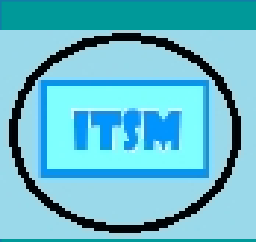
- In the ERP system, input to general ledger occurs simultaneously with business transactions
- Many ERP modules cause transaction data to be entered into general ledger, including:
 - ❖ Sales and Distribution (SD)
 - ❖ Materials Management (MM)
 - ❖ Financial Accounting (FI)
 - ❖ Controlling (CO)
 - ❖ Human Resources (HR)
 - ❖ Asset Management (AM)



Operational Decision-Making Problem: Credit Management



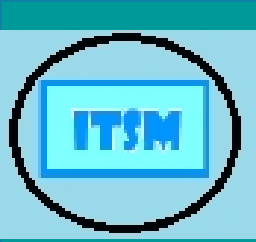
- **Unintegrated information system**
 - ❖ Out-of-date or inaccurate accounting data can cause problems when a company is making operational decisions
 - ❖ Industrial credit management
 - ❖ Fitter credit management procedures
 - ❖ Credit management in SAP ERP



Industrial Credit Management



- **Credit management requires a good balance between:**
 - ❖ Granting sufficient credit to support sales *and ensuring* the company does not lose too much money by granting credit to customers
- **Setting a limit on how much money a customer can owe at any one time**
 - ❖ Monitoring that limit as new orders come in and payments are received



Industrial Credit Management (cont'd.)



- Sales representative needs to be able to review an up-to-date accounts receivable balance when an order comes in
 - Problems arise if Marketing and Accounting have unintegrated information systems
 - ❖ Less than full cooperation on updates
- Problems should not arise with an integrated information system
- ❖ Accounts receivable is immediately updated



Fitter Snacker's Credit Management Procedures

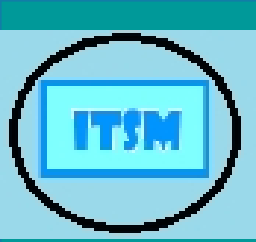
- FS sales clerk refers to a weekly printout of a customer's current balance and credit limit to see if credit should be granted
- Sales data are transferred to Accounting by disk three times a week
- Accounting clerk can use sales input to prepare a customer invoice
- Accounting must make any adjustments for partial shipments before preparing the invoice
- Accounting clerks process customer payments



Credit Management in ERP



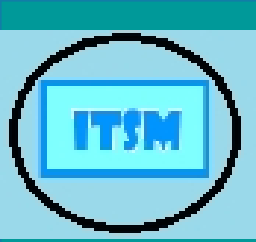
- ERP would allow FS to set a credit limit for each customer
- Company can configure any number of credit-check options in ERP system
- Advantages of using ERP to manage credit
 - ❖ Process is automated
 - ❖ Data are available in real time



Product Profitability Analysis



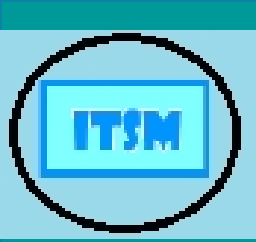
- Business managers use accounting data to perform profitability analyses of a company and its products
- When data are inaccurate or incomplete, the analyses are flawed
- Main reasons for inaccurate or incomplete data
 - ❖ Inconsistent recordkeeping
 - ❖ Inaccurate inventory costing systems
 - ❖ Problems consolidating data from subsidiaries



Inconsistent Recordkeeping



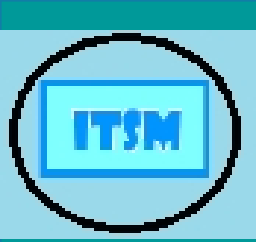
- Each of FS's marketing divisions maintains its own records and keeps track of sales data differently
- Paper records might be inaccurate or missing, making validity of the final report questionable and the creation of the report time consuming
- Without integrated information systems, accounting and reporting to management requires:
 - ❖ Working around limitations of information systems to produce useful output
- ERP system minimizes or eliminates these problems



Inaccurate Inventory Costing Systems



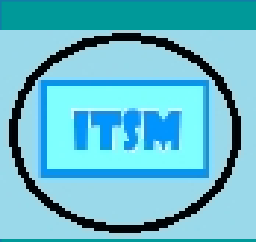
- Correctly calculating inventory costs
 - ❖ One of the most important and challenging accounting tasks in any manufacturing company
- Inventory cost accounting background
 - ❖ Manufactured item's cost has three elements:
 1. Cost of raw materials
 2. Cost of labor employed directly in production of item
 3. all other costs called overhead



Inaccurate Inventory Costing Systems (cont'd.)



- Inventory cost accounting background (cont'd.)
 - ❖ **Direct costs:** materials and labor
 - Can be estimated fairly accurately
 - ❖ **Indirect costs:** overhead items
 - Difficult to associate with specific product(s)
 - ❖ **Standard costs** for a product are established by:
 - Studying historical direct and indirect cost patterns
 - Taking into account the effects of current manufacturing changes
 - ❖ **Cost variances:** differences between actual costs and standard costs



Inaccurate Inventory Costing Systems (cont'd.)



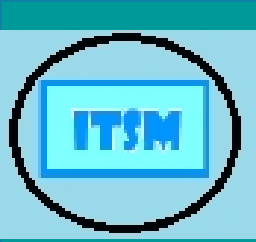
- **ERP and inventory cost accounting**

- ❖ Many companies with unintegrated accounting systems analyze their cost variances infrequently

- Often, they do not know how much it actually costs to produce a unit of a product

- ❖ If FS had an ERP system, employees throughout the company would have recorded costs in a company-wide database as they occurred

- ❖ ERP system configurations allow analysts to track costs using many bases by job , by work area or by production activity.



Inaccurate Inventory Costing Systems (cont'd.)



- **Product costing example**
 - ❖ Suppose Fitter Snacker's wishes to update standard costs for NRG-A bars
 - ❖ Product cost analysis for NRG-A bar
- **Product cost analysis in ERP**
 - ❖ Product cost variant: method for developing a product cost in an ERP system

	NRG-A Bar Product Cost Analysis (7 cases)			
Ingredient	Unit of measure	NRG-A	Cost per unit of measure	Direct material cost
Oats	lb	300	\$0.20	\$60.00
Wheat germ	lb	50	\$0.30	\$15.00
Cinnamon	lb	5	\$3.00	\$15.00
Nutmeg	lb	2	\$4.50	\$9.00
Cloves	lb	1	\$5.50	\$5.50
Honey	gal	10	\$6.40	\$64.00
Canola	gal	7	\$1.70	\$11.90
Vit./min. powder	lb	5	\$18.45	\$92.25
Carob chips	lb	50	\$2.10	\$105.00
Raisins	lb	50	\$3.20	\$160.00
Total direct material cost				\$537.65
Production overhead cost (100% of Total direct material)				\$537.65
Direct labor				54.50
Cost of goods manufactured (COGM)				1,129.80
Sales and administrative costs (30% of COGM)				338.94
Cost of goods sold (COGS)				1,468.74
COGM per case				\$161.40
COGS per case				\$209.82

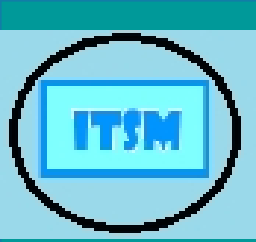
Figure 1. Product cost analysis for NRG-A bar

- **Activity-based costing and ERP**

- ❖ Activity-based costing (ABC)

- Accountants identify activities associated with overhead cost generation and then keep records on *costs and on activities*

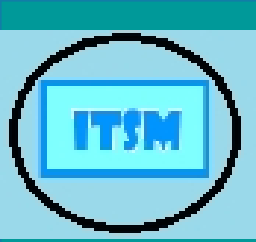
- ❖ ABC requires more bookkeeping than traditional costing methods



Companies with Subsidiaries



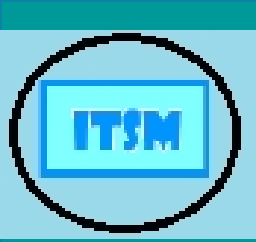
- Account balances for each entity must be compiled and forwarded to the home office
- Consolidated statement for the company as a whole must be created
- Currency translation
 - ❖ Problems when **currency translation** is needed for a subsidiary's accounts
- Intercompany transactions
 - ❖ Transactions that occur between companies and their subsidiaries



Management Reporting with ERP Systems



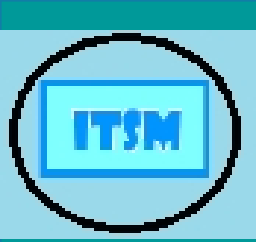
- Generating the right reports for the right situation is often challenging
- Without an ERP system, the job of tracking all the numbers that need to go into a report is a monumental undertaking
- With ERP system, vast amount of information is available for reporting purposes



Document Flow for Customer Service



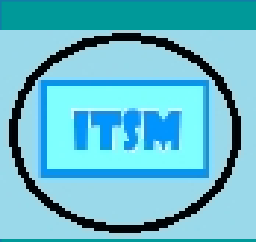
- With an ERP system, all transactions in all areas of a company get posted in a centralized database
- Each transaction posted in ERP gets its own unique document number
 - ❖ Allows quick access to the data
- In ERP, document numbers for related transactions are associated in the database
 - ❖ Provides an electronic audit trail



Built-In Management-Reporting and Analysis Tools



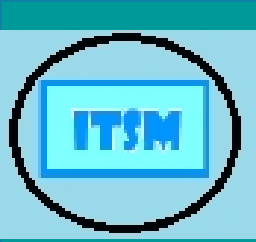
- Accounting records maintained in the common database
- Advantage of using a database is the ability to query the records to:
 - ❖ Produce standard reports
 - ❖ Answer ad hoc questions
- ERP provides a **data warehouse** within each major module
 - ❖ Data warehouse: repository for data from various sources



Archiving



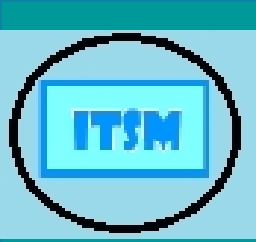
- ERP software offers very few ways to delete items
- Data are removed from ERP system only after they have been recorded to media (tape backup, DVD-R) for permanent storage
- **Archive:** permanent storage
- ERP systems keep track of when data are created or changed
 - ❖ Change Record



User Authorizations



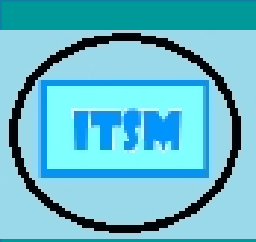
- ERP has sophisticated user administration tools that allow different levels of authorization management
 - ❖ Ensure that employees can perform only the transactions required for their jobs
- Profile Generator
 - ❖ Provides a simple method for selecting functions that a user should be allowed to perform



Tolerance Groups



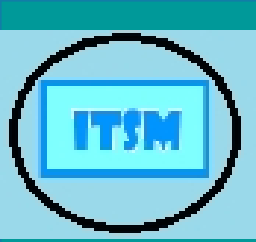
- Setting limits on the size of transaction an employee can process
 - ❖ In an ERP system, this is done using tolerance groups
- Tolerance groups
 - ❖ Preset limits on an employee's ability to post transactions
 - ❖ Set limits on the dollar value for a single item in a document as well as the total value of document



Financial Transparency



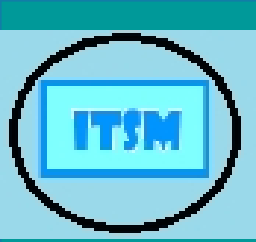
- ERP systems provide the ability to drill down from a report to the source documents (transactions) that created it
 - ❖ Makes it easier for auditors to confirm the integrity of reports
- With a properly configured and managed ERP system, there are direct links between the company's financial statements and individual transactions that make up the statements
 - ❖ Fraud and abuse can be detected more easily



Trends in Financial Reporting (XBRL)



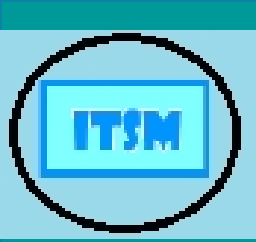
- Extensible Business Reporting Language (XBRL)
 - ❖ Standards based language
 - ❖ Extensible Markup Language (XML) coded data directly from web page into database
 - ❖ Reports processed faster and validated easier
 - ❖ ERP systems accept data in XML and XBRL



Summary



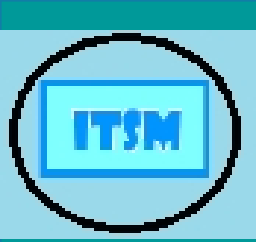
- Companies need accounting systems to record transactions and generate financial statements
- Unintegrated information systems
 - ❖ Accounting data might not be current
 - Can cause problems for sales representatives trying to make operational decisions
 - ❖ Data can be inaccurate
 - Can affect decision making and therefore profitability



Summary (cont'd.)



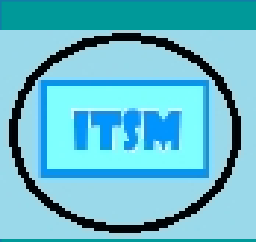
- Closing the books at the end of an accounting period can be difficult with an unintegrated IS, but is relatively easy with an integrated IS
 - ❖ Closing the books means zeroing out temporary accounts
- Using an integrated IS and a common database to record accounting data has important inventory cost-accounting benefits
 - ❖ Can lead to more accurate product cost calculations
 - ❖ Can help managers determine which products are profitable and which are not



Summary (cont'd.)



- Use of an integrated system and a common database to record accounting data has important management-reporting benefits
 - ❖ Built-in drill-down and query tools available
- Trends in financial reporting
 - ❖ XBRL
 - ❖ XML
 - ❖ ERP systems accept data in XML and XBRL into database



Quizz



1. Transactions that occurs between a parent company and one of its subsidiaries.
 - A. Intercompany transactions
 - B. Currency translation
 - C. Financial transparency

2. What system are core software programs used by companies to integrate and coordinate information in every area of the business?
 - A. Unintegrated information system
 - B. Financial information system
 - C. Enterprise Resource Planning